

# POSTPILOT AI

**STUDENT NAME: V. AKASH VARMA**

**REG. NO: 99230041018**

**ASSESSMENT TYPE: MINI ASSESSMENT**

## 1. Task Description

This assessment asked for a multi-step prompt workflow, built and run entirely inside an AI chat interface, with no coding involved. The idea was to take one task and break it into a sequence of clearly separated prompt stages, where whatever comes out of one stage is carried forward on purpose into the next one, instead of starting fresh each time. The point of the exercise was to actually design and chain prompts properly, not to build an app or a tool.

## 2. Chosen Problem Area

Small business owners - a café, a home bakery, a local repair shop, anything with an Instagram page and no marketing person on staff - usually post whatever comes to mind whenever they remember to. There's rarely any real thought given to which idea to post first, whether it actually fits the audience, or whether the final post is even in a shape that's easy to schedule and reuse. Ideas get picked based on gut feeling rather than anything structured, so posts that could've done well get skipped, and posts that don't really fit the audience go up anyway.

This workflow tries to fix that gap with a short, three-stage prompt sequence that takes a plain business description as input and produces one finished, ready-to-schedule Instagram post as output - complete with a caption, hashtags, the content category it belongs to, and a suggested posting time.

## 3. Workflow Architecture

The workflow moves through three stages, in this order. Each stage is given whatever the earlier stage(s) actually produced, so the reasoning builds on real output rather than restarting from the raw business description every time.

- Content Idea Extraction (Few-shot)
- Content Prioritisation (Chain of Thought)
- Final Post Formatting (JSON mode)

## 4. Workflow Stages and Expected Outputs

Content Idea Extraction: pulls a short list of realistic content ideas straight out of the business description, before any decision is made about what to actually post. This produces a bullet list where each idea comes with its content category and a one-line reason it should work.

Content Prioritisation: looks at that list and reasons through which idea makes the most sense to post first, weighing things like effort required, how quickly it builds trust, and how well it matches the audience. This produces a short reasoning trail ending in one clearly chosen idea.

Final Post Formatting: takes the chosen idea and turns it into an actual, structured post - caption, hashtags, content category, and posting time - in a format that could be dropped straight into a scheduling tool without anyone needing to tidy it up first.

## 5. Detailed Prompt Design

The three prompts below were used in order, inside the AI chat window. Each one is written to explicitly pull in the previous stage's real output rather than repeating the original business description from scratch.

### Stage 1: Content Idea Extraction

*"Given a short business description, extract 3 Instagram content ideas as a bullet list with fields: idea, content\_pillar, why\_it\_works. Here are two examples of the format I want:*

*Example 1 - Business: 'A small home bakery specializing in eggless cakes, run by two sisters.'*

- idea: short reel of the sisters decorating a cake together*
- content\_pillar: behind-the-scenes*
- why\_it\_works: shows the personal, family-run angle that big bakeries can't copy*

*Example 2 - Business: 'A neighborhood bike repair shop, known for same-day fixes.'*

- idea: before/after photo of a badly damaged bike getting fixed*
- content\_pillar: product highlight*
- why\_it\_works: visual proof of skill builds trust faster than words*

*Now do the same for this business: [insert business description]"*

### Stage 2: Content Prioritisation

*"Using the content ideas above, think step by step about which one to post first this week. Consider which idea builds trust the fastest, which one is realistic to film with just a phone, and which one best fits an audience that seems to prefer personal, non-promotional content. Walk through your reasoning, then end with one line: Chosen idea: <idea>."*

### Stage 3: Final Post Formatting

*"Using the chosen idea above, write the final Instagram post. Return ONLY valid JSON, nothing else before or after it, in exactly this shape:*  
*{"caption": str, "hashtags": [...], "content\_pillar": str, "best\_posting\_time": str}"*

## 6. Prompt Chaining and Information Flow

What makes this a workflow rather than just three separate questions is that each stage genuinely depends on the one before it. Stage 1 doesn't know or care which idea will eventually get chosen - it just lays out honest options. Stage 2 is where the real thinking happens, because it has to weigh the actual ideas Stage 1 came up with against the audience and effort involved, instead of picking a random one. Stage 3 doesn't get to invent a new idea at this point either - it only has permission to take whatever Stage 2 already decided and turn it into a clean, structured post. If someone questioned the final caption, you could trace it straight back to the chosen idea, and from there back to the original list it came from - nothing in the final output appears out of nowhere.

## 7. Demonstration Example

Business description used: "GreenLeaf Organic Café, a small café in Hyderabad known for millet-based snacks and cold-pressed juices, popular with health-conscious working professionals and college students nearby."

Expected progression through the workflow:

- Stage 1 comes back with three ideas: a behind-the-scenes kitchen reel, a product close-up of the new millet cookies, and a short customer-interview clip.
- Stage 2 reasons that the customer interview needs the least equipment, fits the audience's preference for personal content, and builds trust faster than a product shot - so it's chosen as the one to post first.
- Stage 3 turns that into a finished post: a warm one-line caption about a regular customer, a set of relevant hashtags, the content\_pillar marked as customer story, and a suggested posting time of Friday 7:30 PM.

Actual Stage 3 output from this run:

```
{
  "caption": "Priya has ordered the same juice every Tuesday for two years. We finally asked her why - her answer made our whole week.",
  "hashtags": ["#HyderabadFoodie", "#CafeCulture", "#SupportLocalHyderabad", "#SmallBusinessLove", "#CustomerStory"],
  "content_pillar": "customer story",
  "best_posting_time": "Friday, 7:30 PM"
}
```

## 8. Why This Workflow Was Not Built as a Single Prompt

Asking one big prompt to "write me a good Instagram post for my business" almost always skips straight to a generic caption - it doesn't stop to consider multiple ideas, it doesn't weigh which one is actually worth posting first, and it definitely doesn't come back in a format you could use without editing it by hand. Splitting this into three stages meant each concern got its own dedicated attention: idea generation wasn't rushed, the choice of what to post first had actual reasoning behind it instead of being the model's first guess, and the final formatting step could be held to a strict, checkable structure. It also means if one stage produces something off, only that stage needs fixing - the rest of the chain doesn't have to be thrown out.

## 9. Expected Learning Outcome

Doing this taught me that breaking a task down isn't just about splitting it into smaller pieces - it's about deciding what each piece is actually responsible for. Stage 1 had to stay honest and not jump ahead to a final answer. Stage 2 had to justify its choice using real information instead of just picking something. Stage 3 had to hold itself to a strict format instead of writing however it felt like. I also learned that the way information flows between prompts matters just as much as the prompts themselves - if a later stage doesn't actually use what an earlier one produced, you don't really have a chain, you just have three prompts sitting next to each other.

## 10. Conclusion

This workflow shows that a short, three-stage sequence of prompts, run entirely inside a chat interface with no code, can take a plain business description all the way to a finished, ready-to-post Instagram update - one that's grounded in an actual reasoning trail rather than a lucky first guess. It didn't need an app or a script to do this, just prompts that were designed to hand off the right information to each other in the right order.